MIL-S-45351B(AT)
26 March 1984
SUPERSEDING
MIL-S-45351A(AT)
19 August 1966

MILITARY SPECIFICATION

SEAL, ENGINE SHROUD

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of seals for engine shroud sealing made of treated fiberglass/aramid fabric, stitched over a corrosion resisting wire mesh core.

1.2 Classification. Items described in this specification shall be of the following types as specified (see 6.2).

Type I - Round bulb, with the tail extension of the cover extending radially from the periphery of the core material.

Type II - Rectangular bulb, with the tail extension of the cover parallel to the longer side of the rectangular core.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this documcnt, or by letter.

FSC 2930

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL-C-20079	- Cloth, Glass, Tape, Textile Glass, and Thread Glass.

STANDARDS
FEDERAL

FED-STD-191	- Textile Test Methods.
FED-STD-601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-45662	- Calibration System Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2000	- Rubber Products in Automotive Applications.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish seals which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the seals to be furnished to the Government. All subsequent seals delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified or referenced herein, in applicable specifications, and on applicable drawings or standards.

3.2.1 Thread. The thread used in stitching the plies of the tail shall be wire inserted fiberglass with teflon coating conforming to MIL-C-20079, type III, class III. Minimum breaking strength shall be 20 pounds when tested in accordance with MIL-C-20079.

3.2.2 Cover cloth. Cover cloth shall be aramid/glass wire inserted cloth with silicone rubber on the inside, and coated on the outside with aluminum pigmented silicone compound. The base fabric shall be plain weave aramid/glass cloth. Reinforcing wire shall be Inconel or stainless steel of the 300 series and wire diameter shall be 0.005 $\pm$ 0.0005 inch (in). The cloth weight shall be 1.65 (maximum) pounds per square yard. Warp and filling yarns shall be strands of wire and fiberglass completely covered by aramid fibers. The yarn shall be not less than 50 percent aramid. The wire must not show in the yarn or cloth. The inner and outer coating shall be silicone rubber Grade 5GE605A19B37E026F19Z of ASTM D-2000. The value of Z shall be the minimum adhesion to the base fabric of 10 pounds per inch of width when tested per method 8211 of FED-STD-601.

3.2.3 Cover coatings. When the coatings are integrated into the weave of the cloth so that separation of tensile tongue is impossible, adhesion values are assumed to exceed 15 pounds per inch. The inner and outer coating shall be complete and continuous. No exposed yarns shall be visible through the coating. The coating shall be smooth without blisters or depressions. The cloth after coating shall have the following properties:

a. Thickness: Not less than 0.050 in
b. Weight: 2.90 $\pm$ 0.29 pounds per square yard
c. Tear strength: Not less than 20 pounds (lbs)

3.3 Design and construction. Seals shall be constructed to conform to the dimensions and form shown on the applicable drawings, specifications and standards (see 6.2). These seals shall consist of a core of the shape specified (see 1.2) and a cover (see 3.2.2). The cover shall overlap the core. Its edge shall extend to form the tail.

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3.3.1 Core of the bulb. The core of the bulb shall be constructed of Inconel alloy or corrosion resisting stainless steel of the 300 series with not less than 16% chromium and 6% nickel. The filler shall be inclosed in a tubular mesh cover of the same material. The configuration and consistency of the filler, diameter of wire and size of mesh in filler and mesh cover shall be such as to provide the compression and recovery characteristics specified in 3.4.2, 3.4.2.1 and 3.4.2.2. When specified on the applicable drawings (see 6.2), the mesh cover shall be drawn over the cut ends of the core engaged upon itself in such a way as to form a bound end without distorting the bulb of the seal.

3.3.2 Tail section. The tail section shall be furnished with bolt holes, or with cut-outs, as specified (see 6.2). Sewing shall be of lock stitches and shall count not less than 3 nor more than 5 stitches per running inch. Sewing shall consist of two rows. One stitch line shall be closely adjacent to the core, the other stitch line shall be approximately 1/4 inch in from the edge of the tail. Bolt holes or cut-outs, when specified, shall not penetrate the stitching adjacent to the bulb.

3.3.3 Length. Length of seals without bolt holes or cut-outs shall not exceed 25 feet.

3.3.4 Bulb and tail. Unless otherwise specified by the acquiring activity (see 6.2), dimensions of bulb and tail (overall width) for types I and II seals shall be as specified in table I.

TABLE I. Bulb and tail dimensions.

Type	Nominal bulb diameter (inches)	Tolerance on overall width (bulb plus tail) Dimension specified on applicable drawing
I	3/8 + 1/16	± 1/16
I	1/2 + 1/16	± 1/16
I	5/8 + 1/16	± 1/16
I	1 + 1/8	± 3/32
I	1-1/2 + 1/8	± 1/8
I	1-3/4 + 1/8	± 1/8
	Nominal rectangular bulb size (inches)	
II	3/4 + 1/16 by 1-1/2 + 1/8	+ 1/8

3.4 Performance.

3.4.1 Flame penetration. When tested as specified in 4.6.1, the seal shall withstand a 2000 degrees Fahrenheit (°F) flame for 15 minutes with no penetration of flame beyond the sealing point and no flame remaining in the seal for more than 5 seconds after removal of the flame source.

3.4.2 Load compression. The seal shall compress with application of loads to the bulb section as specified in table II. Type I seals shall meet compression and recovery requirements regardless of the axial direction of

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application of load. The compression of type II seals shall be accomplished with the direction of the force applied parallel to the tail and with the seal held in position tail down in a fixture having a recess 3/4" wide and 1/4" deep with a slot to accomodate tail which may be cut down to 1/2" wide (see figure 1 and 4.6.2).

TABLE II. Load compression.

Type	Dimensions Nominal bulb size	Load - lbs. per linear inch of seal Percent compression (%)			
		20%	33%	40%	60%
I	3/8	2-1/2 to 7-1/2	7 to 12	12 to 28	
I	1/2	2-1/4 to 7-1/4	5-1/4 to 12-1/4	9 to 23	
I	5/8	2 to 7	4-1/2 to 11-1/2	8-1/4 to 17-3/4	
I	1	11 to 20	18.75 to 32	22.75 to 38	
I	1-1/2	2-1/4 to 5-1/4		5-3/4 to 11-3/4	15 to 33
I	1-3/4	2 to 5		5-1/2 to 10-1/2	16 to 30
II	3/4 by 1-1/2			15 to 45	80 to 160

3.4.2.1 Compression recovery 75°F. When tested as specified in 4.6.2.1, at the applicable percentage compression specified in table III, seals shall recover to the percentage of original nominal bulb size of table III after 15 minutes.

TABLE III. Percentage recovery.

Type	Nominal bulb size	Percentage (%)		
		Compression	Recovery @75°F	Recovery @350°F
I	1	33	90	75
I	3/8	33	90	75
I	1/2	33	90	75
I	5/8	33	90	75
I	1-1/2	40	90	80
I	1-3/4	40	90	80
II	3/4 by 1-1/2	40	90	80

3.4.2.2 Compression recovery 350°F. When tested as specified in 4.6.2.2, at the applicable percentage of compression, seals shall recover to the minimum percentage of original nominal bulb size of table III after 15 minutes.

3.4.3 Tear strength. When tested as specified in 4.6.3, the cover shall exhibit a minimum tear strength of 11 pounds before heat test.

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3.5 Marking. Marking for identification shall be in accordance with MIL-STD-130 (see 4.7).

3.6 Workmanship. Workmanship shall be of such quality as to produce a seal free from frayed or ragged edges and other defects which affect serviceability (see 4.8).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.2) in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified (see 6.2) in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials and construction. To determine conformance to 3.2 through 3.3.4, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Control tests (see 4.5.3.1).

4.3 Tests conditions. Unless otherwise specified (see 6.2) herein or in the equipment specification, measurements and tests shall be made at standard ambient conditions. Standard ambient conditions are:

Air temperature	75 ± 5°F
Relative humidity	50 ± 30 percent
Atmospheric pressure	28.5 +2.0 -3.0 in. Mercury

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4.4 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified herein. In either case a minimum of eight samples shall be submitted. Three will be seals 6 inches long of each size and type to be tested. An additional five samples shall be tear strength samples of fabric covers of the size and cut as specified in the applicable test. Approval of the first article samples by the Government shall not relieve the contractor of the obligation to supply items that are fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction samples shall be examined per table IV and inspected per table V.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select the samples specified in 4.4 from the first one hundred linear feet of each size and type of seal and the first lot of fabric covers produced under the production contract for initial production inspection. Initial production units shall be examined per table IV and inspected per table V.

4.4.3 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the seal assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective (see 6.4).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table IV.

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TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability not within tolerance (see 3.3.4).	SIE 1/
102	Not constructed with materials specified in 3.2 through 3.3.2.	Visual
103	Tail section not constructed as specified (see 3.3.3).	Visual
Minor	AQL 2.5% Defective	
201	Dimension not affecting interchangeability not within tolerance (see 3.3.4).	SIE 1/
202	Improper marking (see 3.5).	Visual
203	Faulty workmanship (see 3.6).	Visual
204	Improper packaging (see 5.1).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Control tests, S3 AQL of 6.5. Control tests with an AQL of 6.5 on the bases of percent defective, shall be performed on a sampling selected in accordance with level S3 of MIL-STD-105 as specified in table V.

4.5.3.2 Control tests, 2 samples per 1000 linear feet. Unless otherwise specified (see 6.2), control test samples 6 inches long, shall be selected at the rate of 2 per each 1000 linear feet produced, or 2 in each lot, except that no more than 4 of each size and type shall be selected in any 30 day period. Tear strength samples shall consist of fabric cover only and shall be selected prior to sewing. Samples shall be examined for the defects specified in 4.5.2.2, and subsequently tested as specified in table V.

4.5.4 Failure. Failure of any sample to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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TABLE V. Classification of tests.

Title	Requirment	Test	First article sample no.	Control	
				2/1000 ft.	S3 AQL 6.5
Flame penetration	3.4.1	4.6.1	1		X
Load compression	3.4.2	4.6.2			
Compression recovery 75°F	3.4.2.1	4.6.2.1	2 & 3		X
Compression recovery 350°F	3.4.2.2	4.6.2.2	2	X	
Tear strength	3.4.3	4.6.3	4 through 8 inclusive	X	

4.6 Method of inspection. Unless otherwise specified all test procedures listed are applicable to both type I and type II. Samples shall be selected and tested as specified in 4.4 and in table V.

4.6.1 Flame penetration test. The bulb of the finished seal shall be compressed to 80 percent of its original nominal size between flat steel plates. The exposed bulb (between plates) shall be subjected to a 2000 ± 25°F flame impingement for 15 minutes. During and after testing, the seal shall be examined to determine conformance to 3.4.1.

4.6.2 Load compression. Determine conformance to 3.4.2 by performing the tests of 4.6.2.1 and 4.6.2.2 as specified in table V.

4.6.2.1 Compression and recovery test for 75° ambient temperature. To determine conformance to 3.4.2.1, a 6-inch length of the seal shall be subjected, at room ambient temperature 75 - 5°F to the following tests:

a. Measure and record diameter or height of bulb between parallel metal plates under a load of 0.2 pounds per linear inch of specimen.
b. Compress specimen to the specified amount.
c. Hold under compression at the specified ambient air temperature for 1 minute.
d. Remove specimen from plates and allow to recover for 15 minutes.
e. Measure and record diameter or height of bulb.
f. Calculate the recovery using the following formula:

$$\frac{\text{recorded bulb size} \times 100}{\text{original bulb size}} = \text{percentage recovery}$$

4.6.2.2 Procedure for 350° temperature. To determine conformance to 3.4.2.2, a 6-inch length of the seal shall be subjected to the following test:

a. Measure and record the diameter or height of bulb between parallel metal surfaces when compressed under a load of 0.2 pounds per lineal inch.

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b. Compress specimen to the applicable amount specified in table III.
c. Place assembly in an oven at 350 ± 5°F.
d. Hold assembly in oven for cycling specified in step (h) below.
e. Remove from oven, remove plates, and leave at room ambient temperature for 15 ± 1 minutes, measure specimen as in "a" and record values.
f. Remove specimen from plates and allow specimen to rest for an additional 60 ± 5 minutes at no load.
g. Compress specimen as in step (b) above and begin next phase.
h. Cycles:

First phase - Specimen under compression for 16 ± 1/2 hours at 350 ± 5°F.

Second phase - Specimen under compression for five hours at 350 ± 5°F.

Repeat first and second phases for a total of four cycles (84 hours of continuous cycling) is attained.

i. Remove from oven and leave at room ambient temperature for 15 ± 1 minutes.
j. Remove specimen from compression plates and allow it to recover at no load for 15 ± 1 minutes.
k. Measure as in step (a) above.
l. Calculate recovery using the following formula:

$$\frac{\text{Final recorded bulb size X 100}}{\text{original bulb size}} = \text{percentage recovery}$$

4.6.3 Tear strength test. To determine conformance to 3.4.3, the seals shall be subjected to the test specified in FED-STD-191, method 5134. Samples shall consist of cover only.

4.7 Marking. Determine conformance to 3.5 by visually examining marking in accordance with MIL-STD-130 (see table IV).

4.8 Workmanship. Determine conformance to 3.6 by examining the seal to assure that there are no frayed or ragged edges nor any other defects which would affect serviceability (see table IV).

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. The seals covered by this specification are intended for use in tactical military vehicles for engine shroud seals where high temperature conditions are present.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of seal required (see 1.2).
c. Applicable drawings (see 3.3).
d. Construction of tail section (see 3.3.2).
e. Nominal bulb size and overall width (see 3.3.5).
f. Any special lot size (see 4.5.1.1).
g. Applicable level of packaging, packing and marking (see 5.1).
h. Number of control test samples and specific control tests to be performed if other than specified (see 4.5.3).
i. If first article samples are not required (see 3.1).
j. If responsibility for inspection shall be other than as specified (see 4.1).
k. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
l. If inspection conditions shall be other than as specified (see 4.3).
m. If preproduction inspection is required (see 4.4.1).
n. If initial production inspection is not required (see 4.4.2).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

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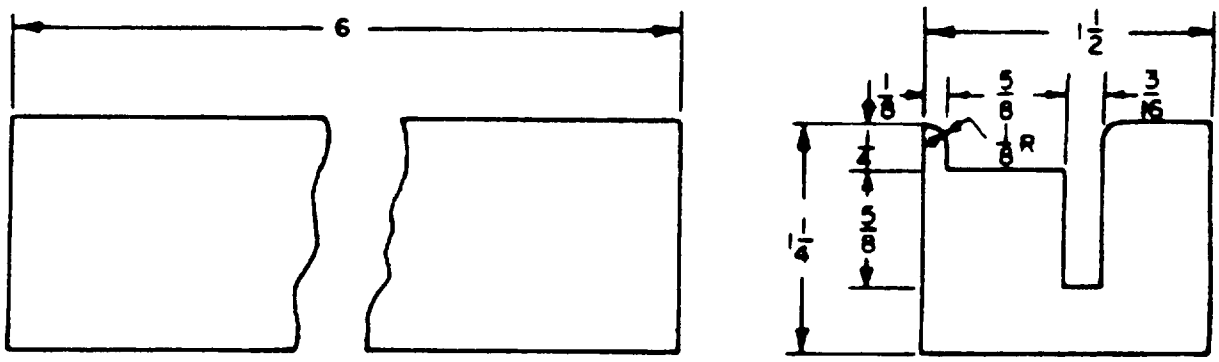


FIGURE 1. Fixture.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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